

Practical-08 Outputs:

memory_allocation.c

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rbhanu_priya@LAPTOP-7PGT... x + v
rbhanu_priya@LAPTOP-7PGTCMPB:~$ cd OSLab
rbhanu_priya@LAPTOP-7PGTCMPB:~/OSLab$ cd fork
rbhanu_priya@LAPTOP-7PGTCMPB:~/OSLab/fork$ nano Practical8.c
rbhanu_priya@LAPTOP-7PGTCMPB:~/OSLab/fork$ gcc Practical8.c -o Practical8
rbhanu_priya@LAPTOP-7PGTCMPB:~/OSLab/fork$ ./Practical8
Before fork:
data[0] = 0
data[99999] = 99999

Child process:
Before modification: data[0] = 0
After modification: data[0] = 999
Child created its own copy due to Copy-on-Write.

Parent process:
data[0] = 0
Parent value is unchanged.
rbhanu_priya@LAPTOP-7PGTCMPB:~/OSLab/fork$ valgrind --leak-check=full ./memory_allocation
valgrind: ./memory_allocation: No such file or directory
rbhanu_priya@LAPTOP-7PGTCMPB:~/OSLab/fork$ valgrind --leak-check=full ./Practical8
==6777== Memcheck, a memory error detector
==6777== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==6777== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==6777== Command: ./Practical8
==6777==
Before fork:
data[0] = 0
data[99999] = 99999

Child process:
```

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Child process:
Before modification: data[0] = 0
After modification: data[0] = 999
Child created its own copy due to Copy-on-Write.
==678==
==678== HEAP SUMMARY:
==678==    in use at exit: 0 bytes in 0 blocks
==678==  total heap usage: 2 allocs, 2 frees, 401,024 bytes allocated
==678==
==678== All heap blocks were freed -- no leaks are possible
==678==
==678== For lists of detected and suppressed errors, rerun with: -s
==678== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)

Parent process:
data[0] = 0
Parent value is unchanged.
==677==
==677== HEAP SUMMARY:
==677==    in use at exit: 0 bytes in 0 blocks
==677==  total heap usage: 2 allocs, 2 frees, 401,024 bytes allocated
==677==
==677== All heap blocks were freed -- no leaks are possible
==677==
==677== For lists of detected and suppressed errors, rerun with: -s
==677== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
rbhanu_priya@LAPTOP-7PGTCMPB:~/OSLab/fork$ |

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Using fork:

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rbhanu_priya@LAPTOP-7PGTCMPB:~$ cd OSLab
rbhanu_priya@LAPTOP-7PGTCMPB:~/OSLab$ cd fork
rbhanu_priya@LAPTOP-7PGTCMPB:~/OSLab/fork$ nano Practical8b.c
rbhanu_priya@LAPTOP-7PGTCMPB:~/OSLab/fork$ gcc Practical8b.c
rbhanu_priya@LAPTOP-7PGTCMPB:~/OSLab/fork$ gcc Practical8b.c -o Practical8b
rbhanu_priya@LAPTOP-7PGTCMPB:~/OSLab/fork$ ./Practical8b
Before fork():
Parent data: 10 10 10 10 10

Child before modification:
Child data: 10 10 10 10 10
Child after modification:
Child data: 100 200 10 10 10

Parent after child modification:
Parent data: 10 10 10 10 10
rbhanu_priya@LAPTOP-7PGTCMPB:~/OSLab/fork$ |

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